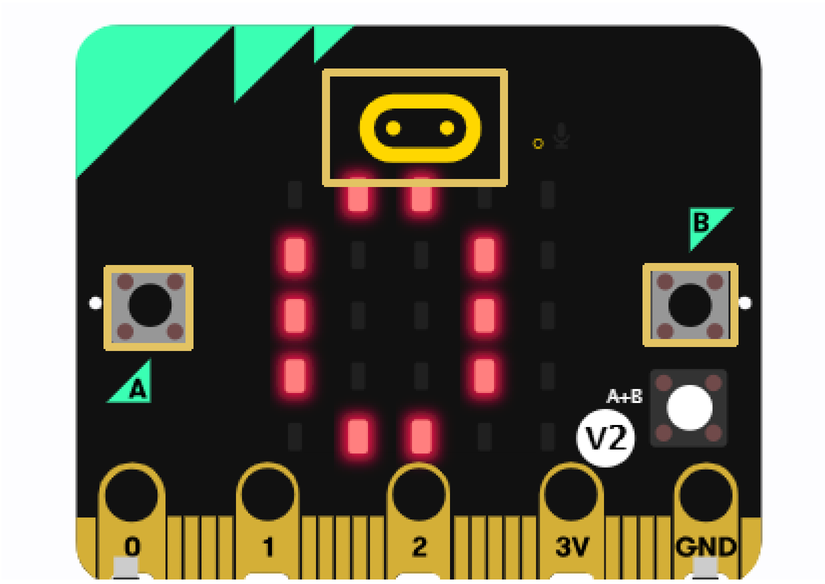
Touch to Sing

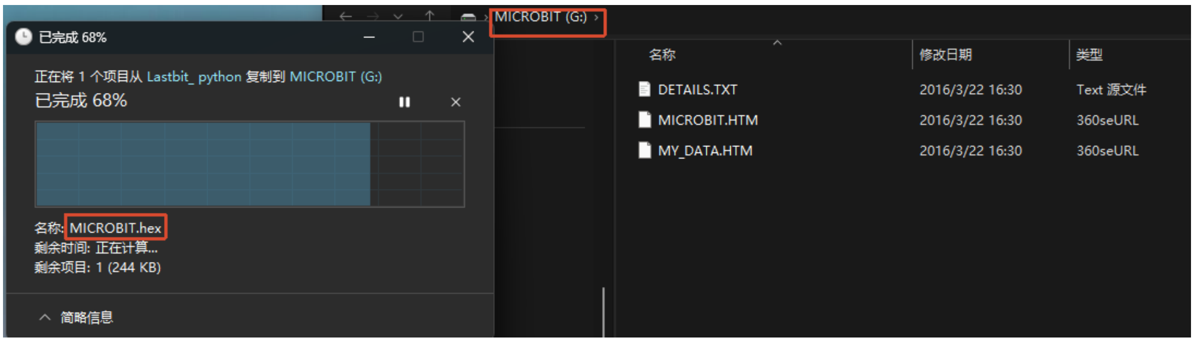
Feature Description

In this section, we will learn how to control the playback of different audio melodies by using the touch Logo and buttons on the Microbit through MicroPython programming.



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



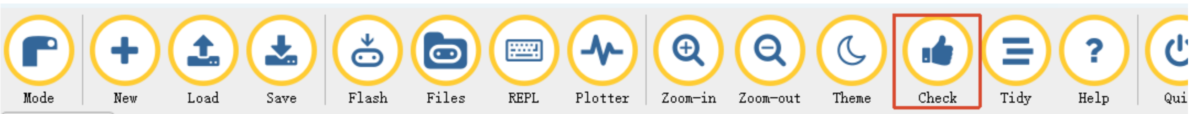
1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.

6. Switch the switch to the ON position.

Code Implementation

```
# 3.Touch_to_Sing

#  Import the microbit and music module
from microbit import *
import music

#  List of melodies
melodies = [
    music.DADADADUM,
    music.ENTERTAINER,
```

```
    music.PRELUDE,
    music.ODE,
    music.BA_DING,
    music.WEDDING,
    music.CHASE
]

TOTAL_MELODIES = len(melodies)
melody_index = 0

def play_current_melody():
    """ Play the current melody"""
    music.stop()
    music.play(melodies[melody_index], wait=False)
    display.show(melody_index)

#  Show the music note icon
display.show(Image.MUSIC_QUAVER)

#  Play the first melody
play_current_melody()

while True:
    #  Play the next melody
    if pin_logo.is_touched():
        melody_index += 1
        if melody_index >= TOTAL_MELODIES:
            melody_index = 0
        play_current_melody()
        sleep(500)  # Prevent multiple presses

    #  Play the previous melody
    if button_a.is_pressed():
        melody_index -= 1
        if melody_index < 0:
            melody_index = TOTAL_MELODIES - 1
        play_current_melody()
        sleep(300)  # Prevent multiple presses

    # the previous melody
    if button_b.is_pressed():
        melody_index -= 1
        if melody_index < 0:
            melody_index = TOTAL_MELODIES - 1
        play_current_melody()
        sleep(300)  # Prevent multiple presses

    # Stop the current melody
    if button_a.is_pressed() and button_b.is_pressed():
        music.stop()
        display.clear()
        sleep(300)  #  Prevent multiple presses

    sleep(10)
```

`display.show` is used to control the images displayed on the micro:bit 5x5 LED matrix. `Image.MUSIC_QUAVER` means displaying the music note icon.

`pin_logo.is_touched()` is used to detect whether the Logo area of the micro:bit is touched. After touching it, you can switch to the next melody.

`music.play` is used to play audio. The first parameter is the audio to be played, and the second parameter indicates whether blocking playback is used. The `wait` parameter is used to control whether to wait for the playback to complete; it defaults to `True`. If changed to `False`, it will return immediately and continue executing the subsequent code. The `loop` parameter is used to control whether to loop the playback; it defaults to `False`. If changed to `True`, the audio will be played in a loop. Common built-in melody parameters include `music.DADADADUM`, `music.ENTERTAINER`, `music.PRELUDE`, `music.ODE`, `music.BIRTHDAY`, `music.WEDDING`, `music.FUNERAL`, `music.PUNCHLINE`, `music.PYTHON`, `music.BADDY`, `music.CHASE`, `music.BA_DING`, `music.WAWAWAWAA`, `music.NYAN`, `music.RINGTONE`, `music.FUNKY` and `music.BLUES`.

Experimental Phenomenon

The micro:bit LED matrix will display the music note icon, and the buzzer will play the currently selected melody. Touch the Logo area to switch to the next melody, press the A button to switch to the previous melody, press the B button to also switch to the previous melody, and press both the A and B buttons at the same time to stop playback.